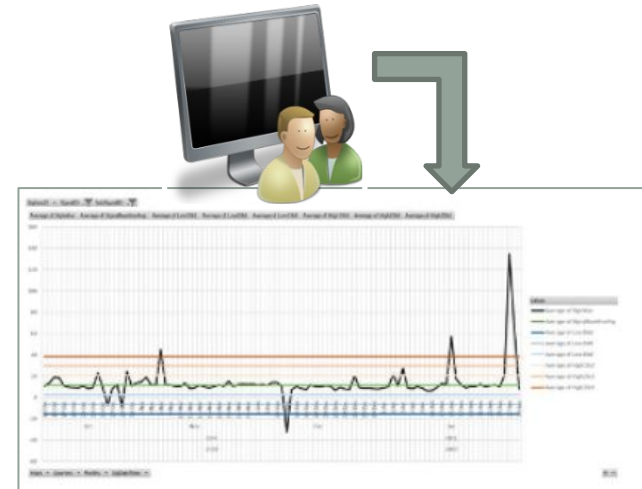


ASPEN CAPABILITIES

- NAS Alarm Notification
- BAHRU Alarm Notification (Prototype)

NAS Alarm Notification: SPC Alarming

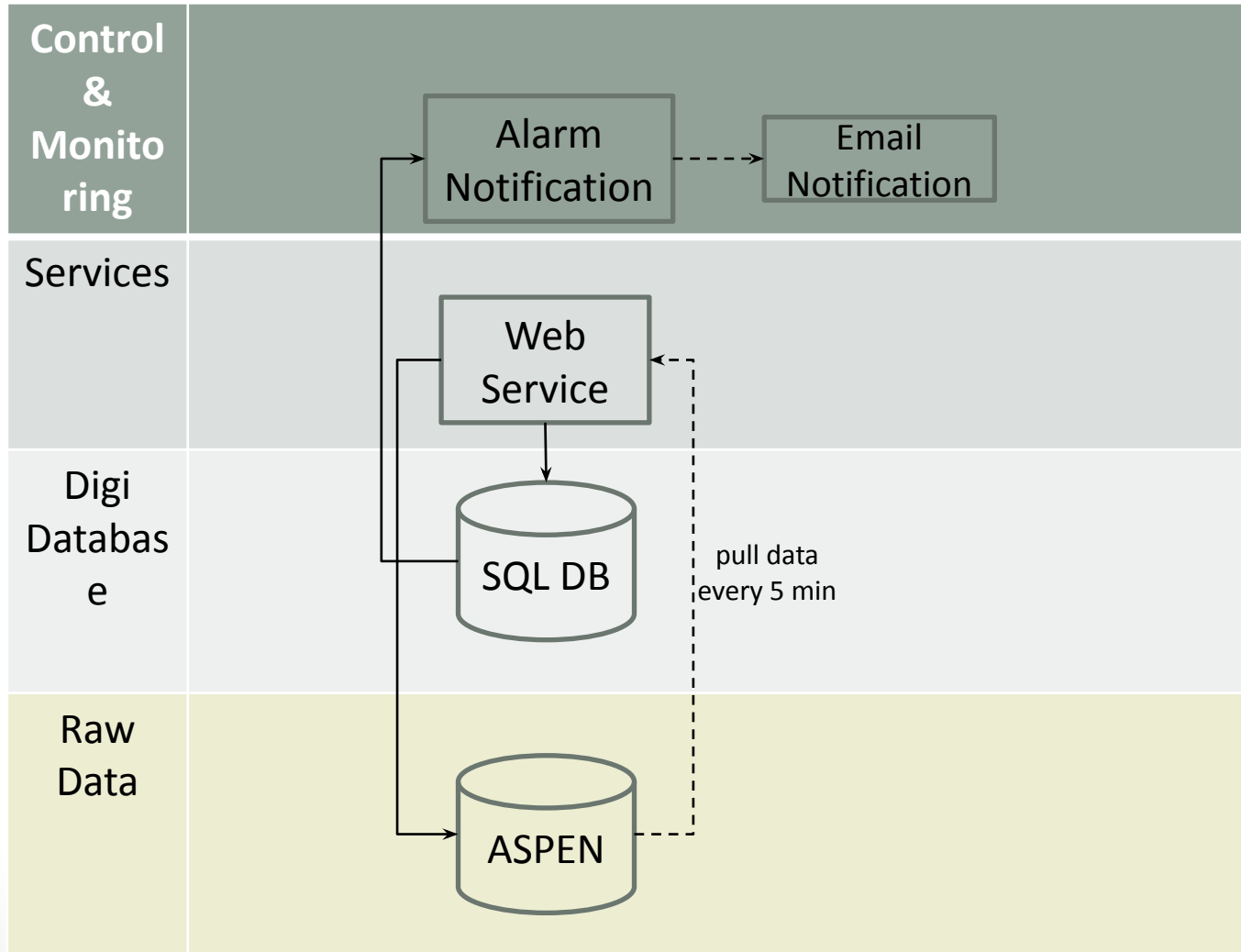
- **Alarming:** Basic threshold alarm system.
- **Tracking:** Tracking coil processing and history.
- **Monitoring:** Engineers used any combination of data sources (MES/Data Warehouse) to create a value (Query able structure).
- **Maintaining:** Digi is working with engineers on training and developing these signals.



AlarmList	Trigger	Common Cause
Major Deviation	1 Data Point Outside of 4 Standard Deviations	Equipment or Sensor Failure
3 Dev	2 of 3 consecutive outside 3 StDev	Sharp Process Change/Failure
2 Dev	3 of 5 consecutive outside 2 StDev	Process Change/Fast Drift
1 Dev	7 of 9 consecutive outside 2 StDev	Process Change/Med Drift
One Side Alarm	13 consecutive points on pos or neg side within range	Slow Drift
Saw Tooth	9 alternating between pos and neg side within range	Overcorrection/Learning by systems

- Allows for complex product based live alarming.
- Detects slow drift and sudden step changes.
- Full control by Process engineers.
- 3 current alarms:
 - Bending, Deformation Curves and Learning monitoring.

BAHRU Alarm Notification (Prototype)



BAHRU Alarm Notification (Prototype)

- **Alarming:**

- Basic threshold alarm system.
- Email notification triggered once the signal passes the threshold.

- **Monitoring:**

- Current data plotted are real-time data.
- By default, it shows all data yet it can be viewed per hourly or selected by date from 7 am to 7 am the next day.

- **Maintaining:**

- A web services pull data every 5 minutes from Aspen.

- **Configurable:**

- The end-user is allowed to :
 - Set min/max value in lookup table.
 - Set email address for email triggering.

- **Limitation:**

- Data availability: Can't used various data sources. Limit only from Aspen.
- No tracking coil processing and history.

Features:

Main Panel:

Process Monitoring | Chart Alarming | Maintenance

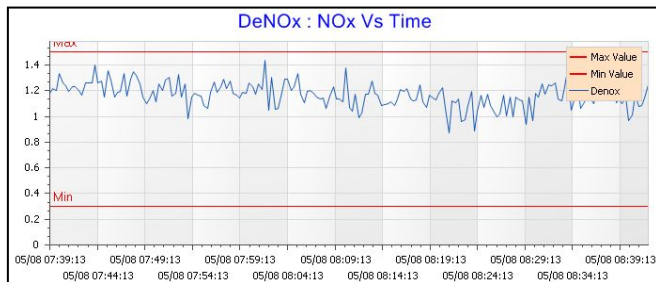
Hourly time / Date selection:

Hours: 1 ☐ Show DateTime Date Time From: Date Time To:

Current coil detail:

CoilID: C0061972 Width: 1260 Length: 5690 Thickness: 0.391 SteelGrade: B430500 SpecialProcCode: APF

Chart:



Lookup table:

#	Steelgrade	Process Code	Thickness From	Thickness To	Comment	Min Value	Max Value	Active
	B430500	APF	0.5	0.4		0.3	1.5	☐

Steelgrade: B430500 Process Code: APF
 Thickness From: 0.5 Thickness To: 0.4
 Comment: Min Value: 0.3
 Max Value: 1.5 Active: ☐

[Update](#) [Cancel](#)

Email setup:

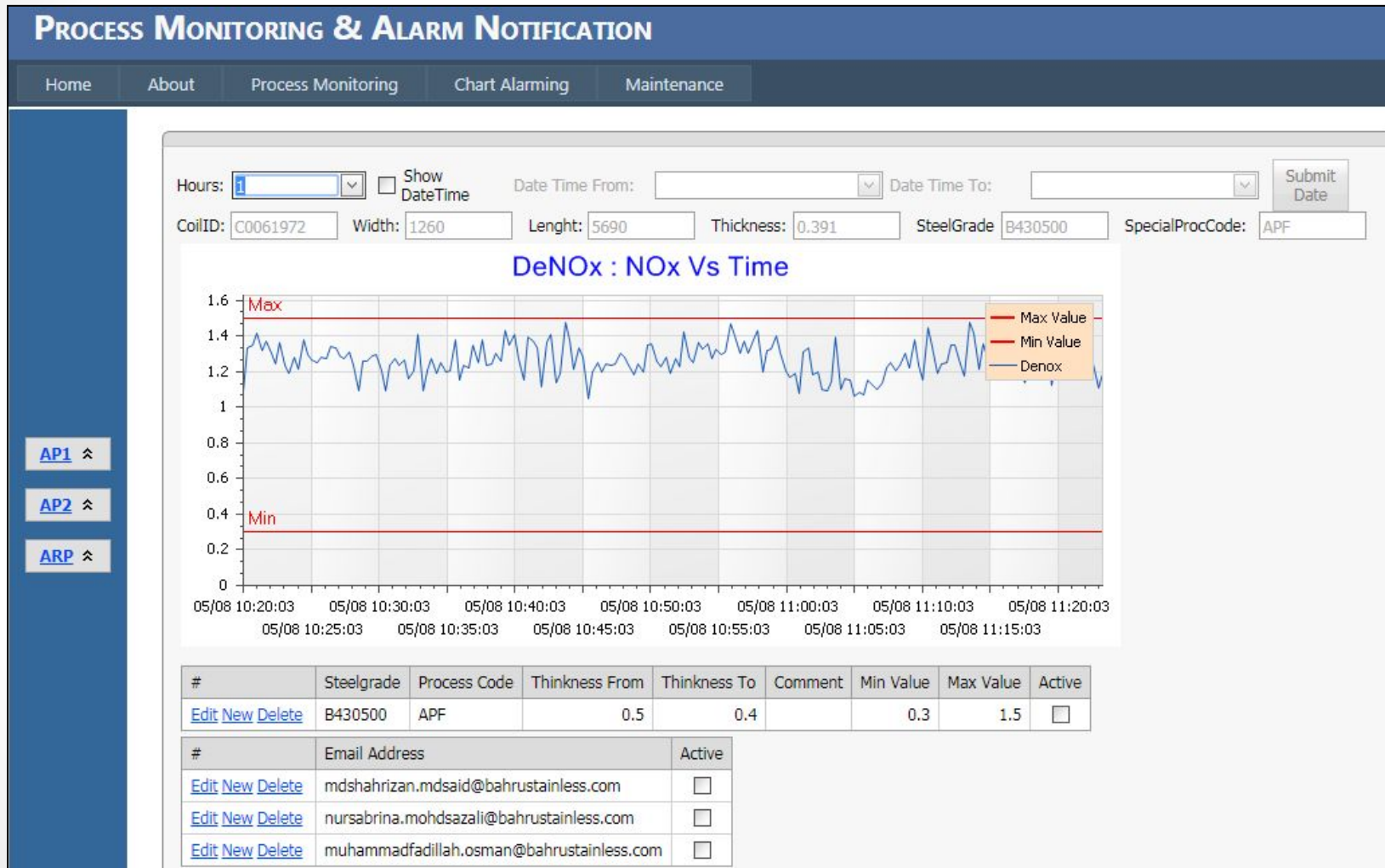
#	Email Address	Active
Edit New Delete	mdshahrizan.mdsaid@bahrustainless.com	☐
	nursabrina.mohdsazali@bahrustainless.com	☐

Email Address: nursabrina.mohdsaza Active: ☐

[Update](#) [Cancel](#)

#	Email Address	Active
Edit New Delete	muhammadfadillah.osman@bahrustainless.com	☐

BAHRU Alarm Notification (Prototype)



<http://172.22.33.45/AlarmNotification/Chart1.aspx>